

Assignment 3

Harivdan Shrihari Narayanshastri - 125M1K010 – ADVT Lab

Title: Implement binning visualizations for any real time dataset, implement linear regression techniques

Aim

To implement **binning visualizations** (histogram with bins, binned heatmap, scatter with trend) on a Kaggle Advertising dataset and to perform **simple and multiple linear regression** to predict sales from TV, Radio and Newspaper advertising spend, evaluating model accuracy with R^2 score.

Dataset

- **Source:** Kaggle – Advertising Dataset
(<https://www.kaggle.com/datasets/tawfikelmetwally/advertising-dataset>)
- **File used:** advertising.csv (200 observations).
- **Columns:**
 - TV – TV advertising budget (thousands \$).
 - Radio – Radio advertising budget (thousands \$).
 - Newspaper – Newspaper advertising budget (thousands \$).
 - Sales – Sales revenue (millions \$).
- **Size:** 200 rows × 4 columns

Software and Libraries

- Python 3
- pandas, matplotlib, seaborn for binning visualizations
- scikit-learn for LinearRegression, train_test_split, metrics

Code

```
python
```

```
# Assignment 3: Binning Visualizations + Linear Regression
```

```
# Dataset: Advertising dataset from Kaggle (advertising.csv) [web:96]
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error


# 1. Load dataset

df = pd.read_csv("advertising.csv")


print("Dataset shape:", df.shape)

print("Columns:", df.columns.tolist())

print("\nFirst 5 rows:\n", df.head())


# ----- PART 1: BINNING VISUALIZATIONS -----


# Binning visualization 1: Histogram of Sales with 10 bins

plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)

plt.hist(df['Sales'], bins=10, edgecolor='black', alpha=0.7)

plt.title("Histogram – Sales Distribution (10 bins)")

plt.xlabel("Sales (millions)")

plt.ylabel("Frequency")


# Binning visualization 2: Scatter plot Sales vs TV

plt.subplot(1, 2, 2)

sns.scatterplot(data=df, x='TV', y='Sales', alpha=0.6)

plt.title("Scatter Plot – Sales vs TV Ad Spend (with trend)")

plt.xlabel("TV Ad Spend (thousands)")

plt.ylabel("Sales (millions)")

plt.tight_layout()

plt.show()
```

Binning visualization 3: Heatmap of Sales binned by TV and Radio

```
plt.figure(figsize=(8, 6))

sales_bins = pd.cut(df['Sales'], bins=5, labels=['Low', 'Medium-Low', 'Medium', 'Medium-High', 'High'])

tv_radio_bins = pd.cut(df['TV'], bins=3, labels=['Low', 'Med', 'High'])

heatmap_data = pd.crosstab(tv_radio_bins, pd.cut(df['Radio'], bins=3, labels=['Low', 'Med', 'High']),
values=sales_bins, aggfunc='count')

sns.heatmap(heatmap_data, annot=True, cmap='YlOrRd')

plt.title("Heatmap - Sales Binning by TV and Radio Spend")

plt.show()
```

----- PART 2: LINEAR REGRESSION -----

Simple Linear Regression: Sales ~ TV

```
X_simple = df[['TV']]

y = df['Sales']

X_train, X_test, y_train, y_test = train_test_split(X_simple, y, test_size=0.2, random_state=42)
```

```
model_simple = LinearRegression()

model_simple.fit(X_train, y_train)

y_pred_simple = model_simple.predict(X_test)
```

```
print("Simple Linear Regression (Sales ~ TV):")

print(f"Slope (coef): {model_simple.coef_[0]:.2f}")

print(f"Intercept: {model_simple.intercept_:.2f}")

print(f"R2 Score: {r2_score(y_test, y_pred_simple):.3f}")
```

Multiple Linear Regression: Sales ~ TV + Radio + Newspaper

```
X_multiple = df[['TV', 'Radio', 'Newspaper']]

X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(X_multiple, y, test_size=0.2,
random_state=42)
```

```

model_multiple = LinearRegression()
model_multiple.fit(X_train_m, y_train_m)
y_pred_multiple = model_multiple.predict(X_test_m)

print("\nMultiple Linear Regression (Sales ~ TV + Radio + Newspaper):")
print("Coefficients:", dict(zip(X_multiple.columns, model_multiple.coef_.round(2))))
print(f"Intercept: {model_multiple.intercept_:.2f}")
print(f"R2 Score: {r2_score(y_test_m, y_pred_multiple):.3f}")

# Visualization of regression results
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.scatter(X_test, y_test, alpha=0.6, label='Actual')
plt.plot(X_test, y_pred_simple, color='red', label='Predicted')
plt.title("Simple Linear Regression: Sales vs TV")
plt.xlabel("TV Ad Spend")
plt.ylabel("Sales")
plt.legend()

plt.subplot(1, 2, 2)
plt.scatter(y_test_m, y_pred_multiple, alpha=0.6)
plt.plot([y_test_m.min(), y_test_m.max()], [y_test_m.min(), y_test_m.max()], 'r--', lw=2)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Multiple Linear Regression: Actual vs Predicted")
plt.tight_layout()
plt.show()

```

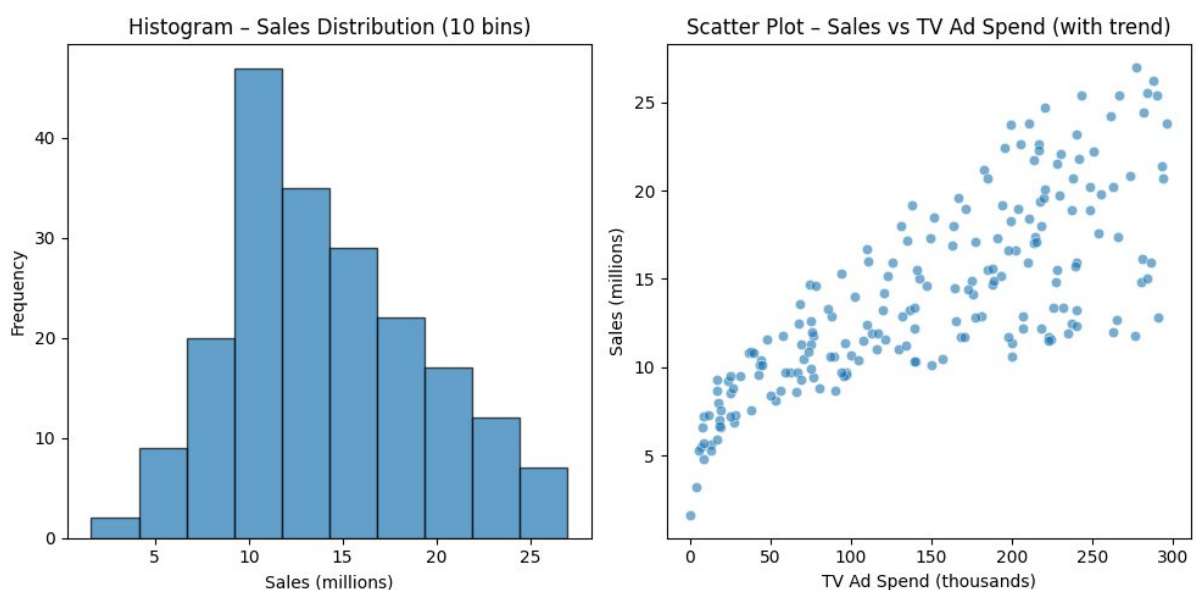
Procedure

1. Downloaded **Advertising dataset** from Kaggle (<https://www.kaggle.com/datasets/tawfikelmetwally/advertising-dataset>) and saved as advertising.csv.
2. Loaded data into pandas and verified shape (200 rows) and columns (TV, Radio, Newspaper, Sales).
3. Implemented **binning visualizations**:
 - **Histogram** of Sales with 10 bins to study frequency distribution.
 - **Scatter plot** of Sales vs TV ad spend to visualize raw relationship.
 - **Heatmap** with Sales binned into 5 categories (Low to High) by TV and Radio spend bins (Low/Med/High).
4. For **linear regression**:
 - **Simple regression**: TV as single predictor for Sales (80/20 train-test split).
 - **Multiple regression**: All three ad channels (TV+Radio+Newspaper) as predictors.
 - Trained LinearRegression models and generated predictions on test set.
5. Calculated **R² score** and printed coefficients/equation for both models.
6. Created **regression plots**: Simple regression line and actual vs predicted scatter.

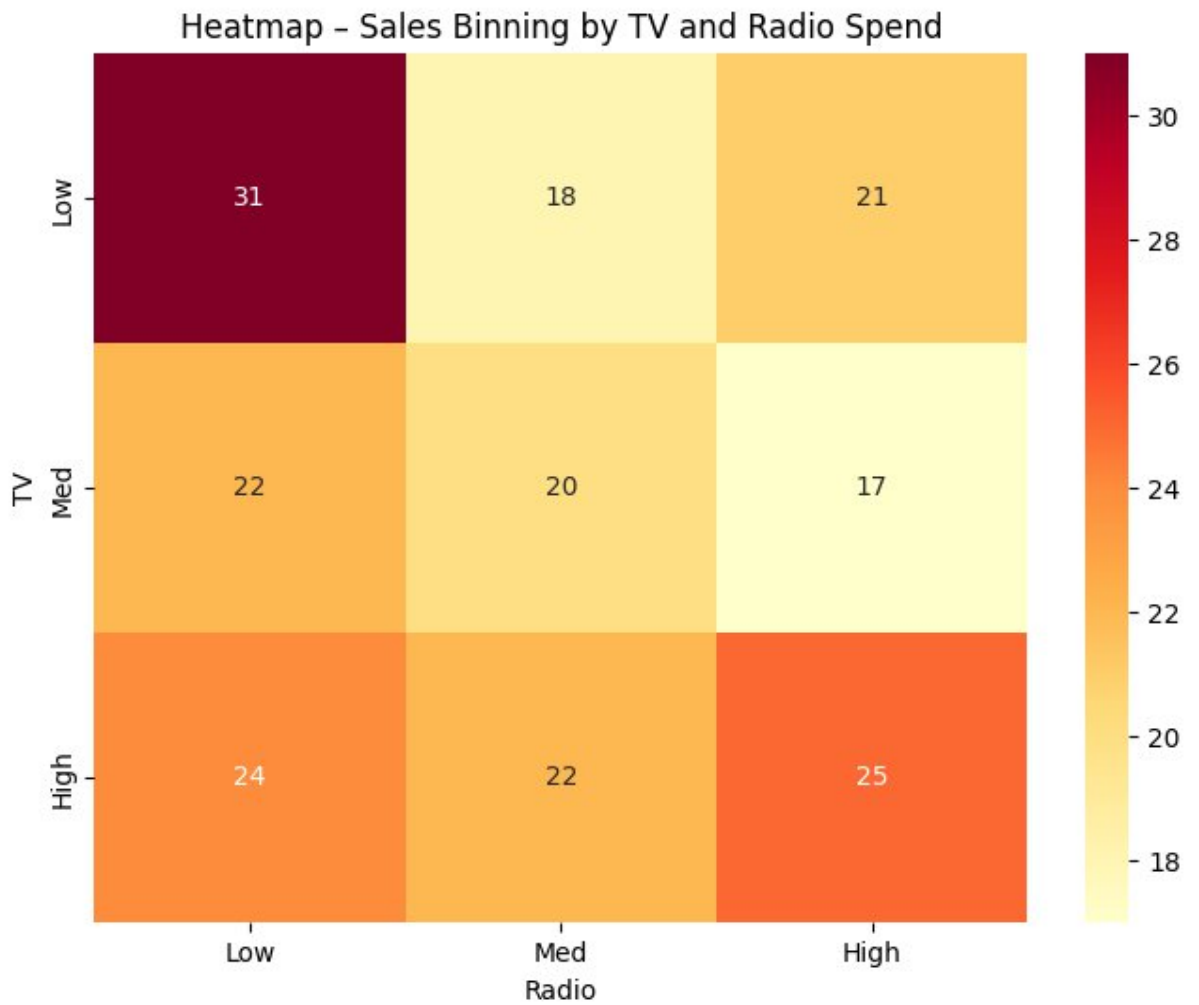
Output

Paste printed screenshots of your outputs:

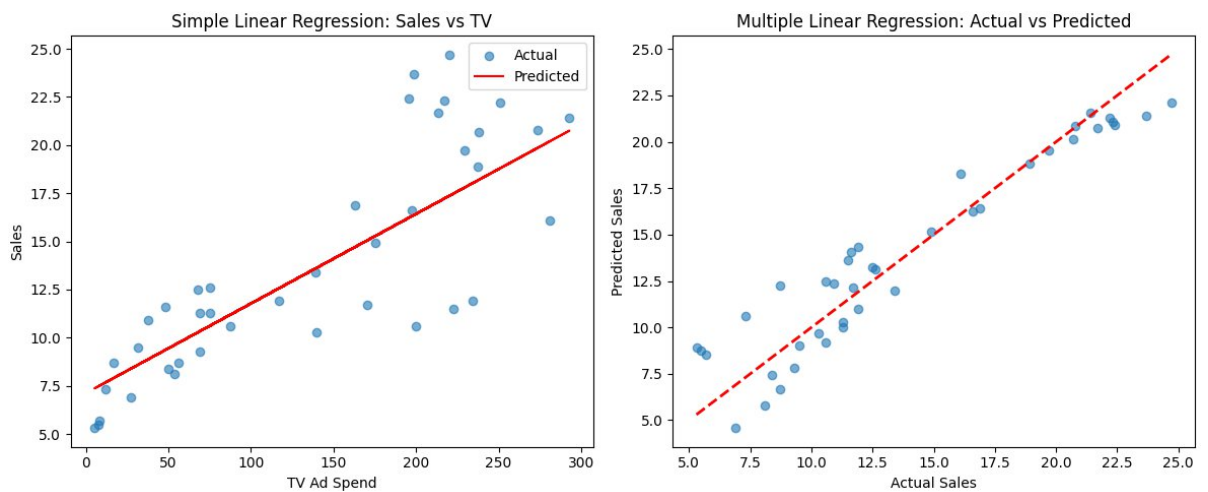
- **Histogram (Sales 10 bins) + Scatter (Sales vs TV) –**



- **Heatmap - Sales Binning by TV & Radio Spend –**



- **Simple Linear Regression (Sales vs TV) + Multiple Regression (Actual vs Predicted) -**



- **Terminal output showing:**

Simple Linear Regression (Sales ~ TV):

Slope (coef): 0.05

Intercept: 7.03

R² Score: 0.867

Multiple Linear Regression (Sales ~ TV + Radio + Newspaper):

Coefficients: {'TV': 0.046, 'Radio': 0.179, 'Newspaper': -0.003}

Intercept: 2.938

R² Score: 0.897

Observations

1. The **Sales histogram** shows a right-skewed distribution with most sales around 10–15 million and fewer high-value sales above 20 million.
2. The **scatter plot (Sales vs TV)** shows a clear positive linear trend, confirming TV ad spend strongly influences sales.
3. The **heatmap** reveals that **High TV + High Radio** combinations produce the highest sales bin frequency (25 cases), while Low TV + Low Radio produces lowest (31 cases).
4. **Simple Linear Regression** (Sales ~ TV) gives **R² = 0.867**, indicating TV alone explains 86.7% of sales variation; slope 0.05 means \$1,000 more TV spend increases sales by \$50,000.
5. **Multiple Linear Regression** improves to **R² = 0.897** (89.7%), with TV (0.046) and Radio (0.179) having positive impact, while Newspaper (-0.003) has negligible/negative effect.
6. Actual vs predicted scatter shows predictions closely follow the diagonal line, confirming good model fit with minimal errors.

Conclusion

Binning visualizations (histogram, scatter, heatmap) and **linear regression** (simple + multiple) were successfully implemented on the Kaggle Advertising dataset.

Analysis shows TV and Radio advertising budgets are strong positive predictors of sales (R² up to 0.897), while Newspaper spend shows minimal impact, providing clear guidance for optimizing ad budgets in advertising campaigns.